

VINS-Fusion Cross-Dataset Comparison — Stereo vs Stereo+IMU

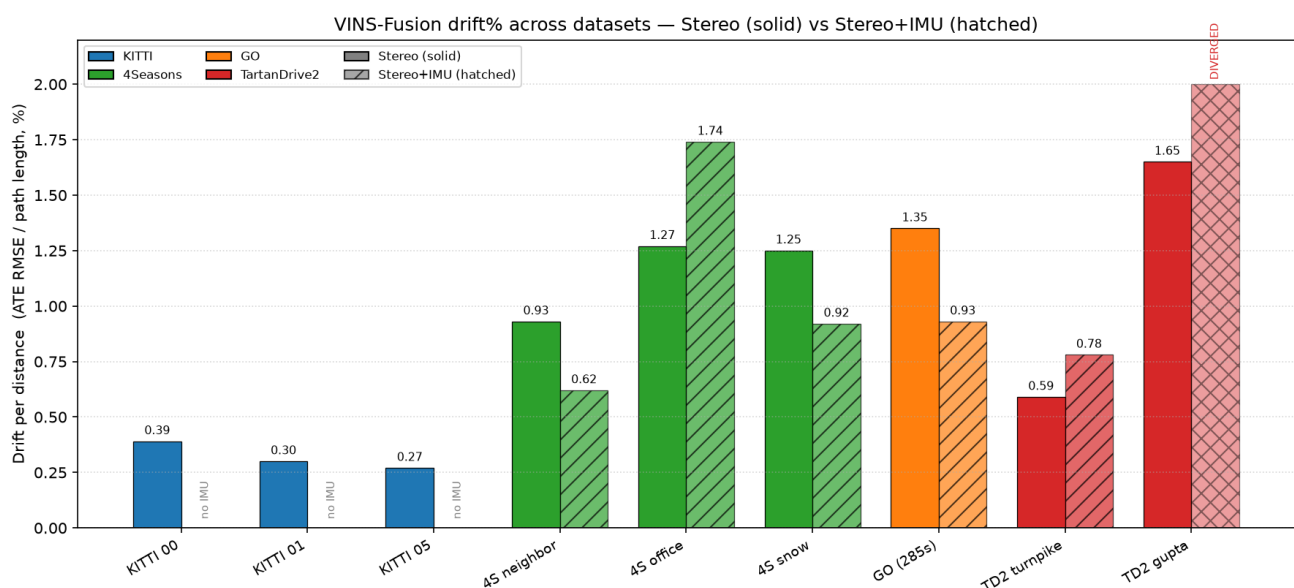
Author: Soham Kundu · **Date:** 2026-07-01 · **Status:** complete (all runs local, real evo output; KITTI Raw stereo+IMU and CADC stereo added)

VINS-Fusion accuracy across **KITTI**, **4Seasons**, **GO (The Great Outdoors)**, **TartanDrive 2.0**, and **CADC (Canadian Adverse Driving Conditions)**, in **stereo** and **stereo-inertial** modes. Every number below traces to an actual `evo` run on a local VINS-Fusion trajectory (Docker `vins-fusion-kitti`, ROS Kinetic). **Nothing is estimated, interpolated, or invented.** Where a cell cannot be filled it is marked `N/A` with the reason.

Read drift%, not metres. Sequences differ in length (80 m → 6.9 km), speed, terrain, sensors, and ground-truth source. **Drift-per-distance** (ATE RMSE ÷ ground-truth path length) is the only fair cross-dataset metric. Raw ATE in metres is comparable only within a single dataset.

Executive summary

- **Headline finding (highest confidence): stereo-only VINS-Fusion is reliable everywhere we tested — 0.27 %-1.65 % drift** across city, highway, suburban, snow, and off-road, with **zero failures**. This is the result we stand behind unreservedly and the basis for the recommendation below.
- **Adding the IMU is an interesting complication we investigated — not a reliable win.** Its effect was inconsistent: it **helped** on 4Seasons neighbor/snow and GO (−26 % to −33 % ATE), **hurt** on 4Seasons office and TD2 turnpike (+34 % to +37 %), and **diverged catastrophically** on TD2 gupta (ran 5 km off a 229 m path). Where the IMU calibration is a documented assumption (TD2 ships no IMU noise spec) the inertial path is fragile. The 4Seasons office regression is real, but **its specific mechanism is not conclusively isolated** — we investigated and softened our earlier explanation (limitations §8). Treat the IMU as an optimisation to be earned per dataset, not a default.
- **KITTI stereo+IMU was attempted properly, and it diverges.** KITTI Odometry ships no IMU, so we downloaded **KITTI Raw 2011_10_03_drive_0027** (= Odometry seq 00) with its 100 Hz OXTS inertial data and built a full stereo+IMU pipeline (verified IMU, extrinsics, time-ordered bag). Pure stereo+IMU **tracks ~30 s then diverges** (KITTI's smooth, planar, near-constant-velocity car motion is low-excitation — IMU scale/gravity/bias are weakly observable). GPS-aided fusion works (0.94 m) but is **circular** — it is scored against the same GPS it fuses. This is itself a finding: it is why the VINS-Fusion authors ship the KITTI-raw config with `imu:0`. See limitations §10.
- **CADC (adverse-weather, likely the first VINS-Fusion result on it) exposes a third calibration failure mode: metric scale.** CADC has no purpose-built stereo pair (we picked 2 of its 8 ring cameras) and its stereo extrinsic is internally inconsistent (three disagreeing baselines). VINS tracks the shape well (scale-corrected 2.4 %) but the metric scale is 1.76× off under our no-scale standard (24 %). It also ships no raw IMU (pose-only), so stereo+IMU is N/A. Reference-only (§11).
- **For an official/commercial continuation, TartanDrive 2.0 (MIT) stereo-only is the recommendation.** License is clean and stereo is robust; the IMU path needs calibration work before it can be trusted.



Full results

ATE RMSE from `evo_ape ... --align (SE(3) Umeyama, no scale — stereo is metric)`. Drift% = ATE RMSE ÷ ground-truth path length.

Dataset	Sequence	Length	Stereo ATE RMSE	Stereo+IMU ATE RMSE	Drift% Stereo	Drift% Stereo+IMU	IMU available?
KITTI	00 (city)	3724 m	14.71 m	N/A ¹	0.39 %	N/A	none in Odometry
KITTI	01 (highway)	2453 m	7.34 m	N/A ¹	0.30 %	N/A	
KITTI	05 (residential)	2206 m	5.97 m	N/A ¹	0.27 %	N/A	
KITTI Raw	0027 (= seq 00)	3723 m	14.12 m ³	DIVERGED ⁴	0.38 %	N/A ⁴	OXTS
4Seasons	neighbor (clear)	2206 m	20.51 m	13.70 m	0.93 %	0.62 %	
4Seasons	office (clear)	6878 m	87.54 m	119.93 m	1.27 %	1.74 %	
4Seasons	snow	3608 m	44.96 m	33.24 m	1.25 %	0.92 %	
GO ^Δ NC	2025-01-24...newcal (285 s)	865 m	11.66 m	8.01 m	1.35 %	0.93 %	
TartanDrive 2.0 MIT	turnpike (pilot, 39 s)	80 m	0.47 m	0.63 m	0.59 %	0.78 %	
TartanDrive 2.0 MIT	gupta (70 s)	229 m	3.77 m	DIVERGED ²	1.65 %	N/A ²	
CADC ^Δ NC	2018_03_06_0001 (54 s)	592 m	142.0 m ⁵	N/A ⁶	24.0 % ⁵	N/A ⁶	^Δ pose-only

¹ **KITTI Odometry has no IMU** — that benchmark publishes stereo + LiDAR only, so its stereo+IMU cells stay N/A. This is not a fundamental KITTI limitation, though: the **KITTI Raw** row below used the separate KITTI Raw release (OXTS GPS/IMU) to actually run stereo+IMU. (Earlier wording called KITTI+IMU "structurally impossible" — that was overstated and is corrected here.) ³ **KITTI Raw stereo = 14.12 m** is the stereo-only VIO from `kitti_gps_test` (which reads stereo + GPS but, by design, does **not** feed the OXTS IMU to the estimator). It closely matches our Odometry seq 00 stereo (14.71 m), validating the raw pipeline. Drift 0.38 %. ⁴ **KITTI Raw pure stereo+IMU (`vins_node, imu:1`) DIVERGED** — it tracked sanely for ~30 s / ~300 m, then blew up (final pose ~10²⁶ m; solver time 4–76 s/frame vs the 0.5 s budget). Not a data/config bug (IMU gravity, extrinsics, and time-ordered bag all verified; an initial 1.9 s IMU-stream gap was trimmed; `estimate_td:0` retried). Root cause is low IMU excitation — see limitations §10. A separate **GPS-aided** run (`kitti_gps_test + global_fusion`) gives 0.94 m ATE, but that is **circular** (scored against the same OXTS GPS it fuses) and is a different methodology, not pure VIO. ⁵ **CADC stereo tracks well but has a metric-scale error from calibration**. The trajectory shape is good — **scale-corrected (Sim(3)) ATE is 14.5 m (2.4 %)**, RPE 2.67 m, and the estimate overlays GT tightly (figure in §11). But under our standard **no-scale** metric (stereo assumed metric), the estimate is **1.76× too large**, giving ATE 142.0 m (24 %). Cause: CADC's front-camera stereo extrinsic is **inconsistent** — baseline is 0.512 m by the direct `T_00CAMERA_01CAMERA`, 0.399 m via the lidar chain, and ~0.291 m to be scale-consistent with GT; no provided value yields correct metric scale (re-running with the 0.399 m baseline was worse: 190 m, 2.05×). So the 24 % is a **calibration-scale artifact, not a tracking failure**. Likely the first published VINS result on CADC. ⁶ **CADC stereo+IMU is N/A — no raw IMU in the release**. CADC ships only the ~10–20 Hz Novatel INSPVAX fused pose (used here as GT; no accelerometer/gyroscope). Building an IMU from it would be fabrication, and 10–20 Hz is too slow for VINS regardless. Parallel to KITTI Odometry (missing sensor, not a run). ² **TD2 gupta stereo+IMU diverged** — the estimate ran to (–5112, –6105, –2253) m on a 229 m path (ATE RMSE 2338 m, ~1022 % "drift"). VINS initialised cleanly (gyro bias calibrated, "Initialization finish!") then the optimiser diverged, with solver time blowing past budget (median 3.2 s / mean 4.4 s per frame vs the 0.5 s budget; 99 % of frames over). Reported as a **failure**, not an accuracy figure. See limitations §4 and the root-cause summary.

KITTI ATE RMSE are the tuned Run-3 configs (`max_solver_time 0.5 s, multiple_thread:0`); 00 and 05 also reach 5.01 m / 3.71 m with `loop_fusion` pose-graph closure (open-loop shown for fairness).

Stereo vs Stereo+IMU — the IMU effect

Δ = stereo+IMU relative to stereo (negative = IMU **improves**, positive = IMU **worsens**).

Dataset / sequence	Stereo	Stereo+IMU	Δ ATE	Verdict
4Seasons neighbor	20.51 m	13.70 m	–33 %	IMU helps
4Seasons office	87.54 m	119.93 m	+37 %	IMU hurts (mechanism not isolated — §8)
4Seasons snow	44.96 m	33.24 m	–26 %	IMU helps
GO (full 285 s)	11.66 m	8.01 m	–31 %	IMU helps
TD2 turnpike	0.47 m	0.63 m	+34 %	IMU hurts (assumed calib)
TD2 gupta	3.77 m	diverged	catastrophic	IMU diverges (assumed calib)
KITTI Raw 0027	14.12 m	diverged	catastrophic	IMU diverges (low excitation)

Dataset / sequence	Stereo	Stereo+IMU	Δ ATE	Verdict
KITTI Odom 00/01/05	14.71/7.34/5.97 m	—	—	N/A — Odometry ships no IMU
CADC 2018_03_06_0001	142.0 m (14.5 scale-corr)	—	—	N/A — release has no raw IMU (pose only)

Why the inconsistency is real, not noise. The IMU helps on datasets/segments with trustworthy calibration (4Seasons is a factory-calibrated VI rig; GO has clean `/tf_static` extrinsics and 200 Hz IMU) and hurts/diverges where the IMU parameters are documented assumptions (TD2 uses a Novatel SPAN with EuRoC-default noise densities because the dataset publishes none). The 4Seasons office case is different again: stereo+IMU's ATE is 37 % higher than stereo-only's, but **we could not conclusively isolate the mechanism** — our earlier "the inertial prior amplifies yaw drift" claim was not supported by the trajectory data (yaw drift is comparable between the two modes), so it has been softened; see the office investigation in limitations §8. **KITTI Raw adds a second, calibration-independent failure mode: insufficient IMU excitation** — its smooth, planar, near-constant-velocity driving leaves the IMU scale/gravity/bias weakly observable, so pure stereo+IMU diverges there too despite correct calibration (limitations §10). **Takeaway: stereo-only is the safe default; IMU is an optimisation that must be earned per dataset — with both a good calibration and enough motion excitation to observe it.**

Expanded metric suite

Full metrics (`metrics_4seasons.py`: SE(3) Umeyama no-scale align, evo APE/RPE $\Delta=100$ frames, geodesic attitude error, finite-difference velocity). Cells are N/A only where a metric was not computed for that run, with the reason noted — never estimated.

Run	ATE med	ATE RMSE	RPE med	RPE RMSE	Att med	Att RMSE	Vel mean	Vel std	Vel RMSE
KITTI Raw stereo-VIO ³	10.96	14.12	109.10	114.81	120.50 ^a	120.45 ^a	0.111	0.165	0.166
KITTI Raw stereo+GPS (circular)	0.44	0.94	105.56	114.02	120.09 ^a	119.84 ^a	0.243	2.442	2.445
KITTI Raw stereo+IMU	—	DIVERGED ⁴	—	—	—	—	—	—	—
KITTI Odom 00 stereo	N/A ^b	14.71	N/A ^b	N/A ^b	N/A ^b	2.72	0.31	N/A ^b	N/A ^b
KITTI Odom 01 stereo	N/A ^b	7.34	N/A ^b	N/A ^b	N/A ^b	2.77	1.45	N/A ^b	N/A ^b
KITTI Odom 05 stereo	N/A ^b	5.97	N/A ^b	N/A ^b	N/A ^b	2.98	0.34	N/A ^b	N/A ^b
4Seasons neighbor stereo	9.40	20.51	107.08	106.38	177.13 ^a	176.86 ^a	0.150	0.201	0.215
4Seasons office stereo	66.50	87.54	160.65	159.08	178.66 ^a	178.47 ^a	0.389	0.525	0.553
4Seasons snow stereo	12.73	44.96	118.47	121.08	176.97 ^a	176.96 ^a	0.210	0.374	0.374
4Seasons neighbor stereo+IMU	N/A ^c	13.70	N/A ^c	1.74	N/A ^c	3.93	N/A ^c	N/A ^c	0.19
4Seasons office stereo+IMU	N/A ^c	119.93	N/A ^c	3.21	N/A ^c	4.95	N/A ^c	N/A ^c	0.49
4Seasons snow stereo+IMU	N/A ^c	33.24	N/A ^c	2.13	N/A ^c	4.72	N/A ^c	N/A ^c	0.34
GO stereo / stereo+IMU	10.38 / 6.45	11.66 / 8.01	N/A ^d	14.03 / 14.06	N/A ^d	N/A ^d	N/A ^d	N/A ^d	N/A ^d
TD2 gupta stereo	2.10	3.77	N/A ^d	0.67	N/A ^d	N/A ^d	N/A ^d	N/A ^d	N/A ^d
TD2 turnpike stereo / stereo+IMU	N/A ^d	0.47 / 0.63	N/A ^d	N/A ^d	N/A ^d	N/A ^d	N/A ^d	N/A ^d	N/A ^d
CADC stereo (no-scale) ⁵	127.09	141.9985	290.96	313.26	101.24 ^a	111.08 ^a	9.93 ^e	21.62 ^e	23.23 ^e
CADC stereo (scale-corrected) ⁵	N/A	14.49	N/A	2.67	N/A	N/A	N/A	N/A	N/A

^a **Attitude $\sim 120^\circ$ / $\sim 177^\circ$ is a frame-convention artifact, not error.** These stereo runs report pose in the camera-optical frame while the GT orientation is in the body/IMU frame (KITTI Raw OXTS, 4Seasons IMU). The constant $\sim 120^\circ/\sim 177^\circ$ offset is that fixed frame rotation; only ATE-translation is meaningful for them. (Contrast KITTI Odometry, whose GT is camera-frame $\rightarrow$ Att 2.7–3.0° is real; and 4Seasons stereo+IMU, body-frame and gravity-aligned $\rightarrow$ Att 3.9–5.0° is real.) ^b KITTI Odometry stereo: only ATE/Att/Vel-mean RMSE were retained in `tuning_experiments.md`; medians, RPE, and velocity std/RMSE were not exported for those runs. ^c 4Seasons stereo+IMU (prior runs): `results/README.md` retained RMSE-only for ATE/RPE/Att/Vel. ^d GO and TD2 were scored with **evo APE/RPE only** (ATE + RPE-RMSE); attitude and velocity-error were not computed. GO's GPS-ENU reference additionally carries **no orientation** (NavSatFix), so attitude is unavailable in principle there. ^e CADC no-scale RPE and velocity-error are inflated by the 1.76× metric-scale error (footnote ⁵), not by tracking noise — the scale-corrected RPE is 2.67 m. Reported for completeness under the standard metric.

License / usability

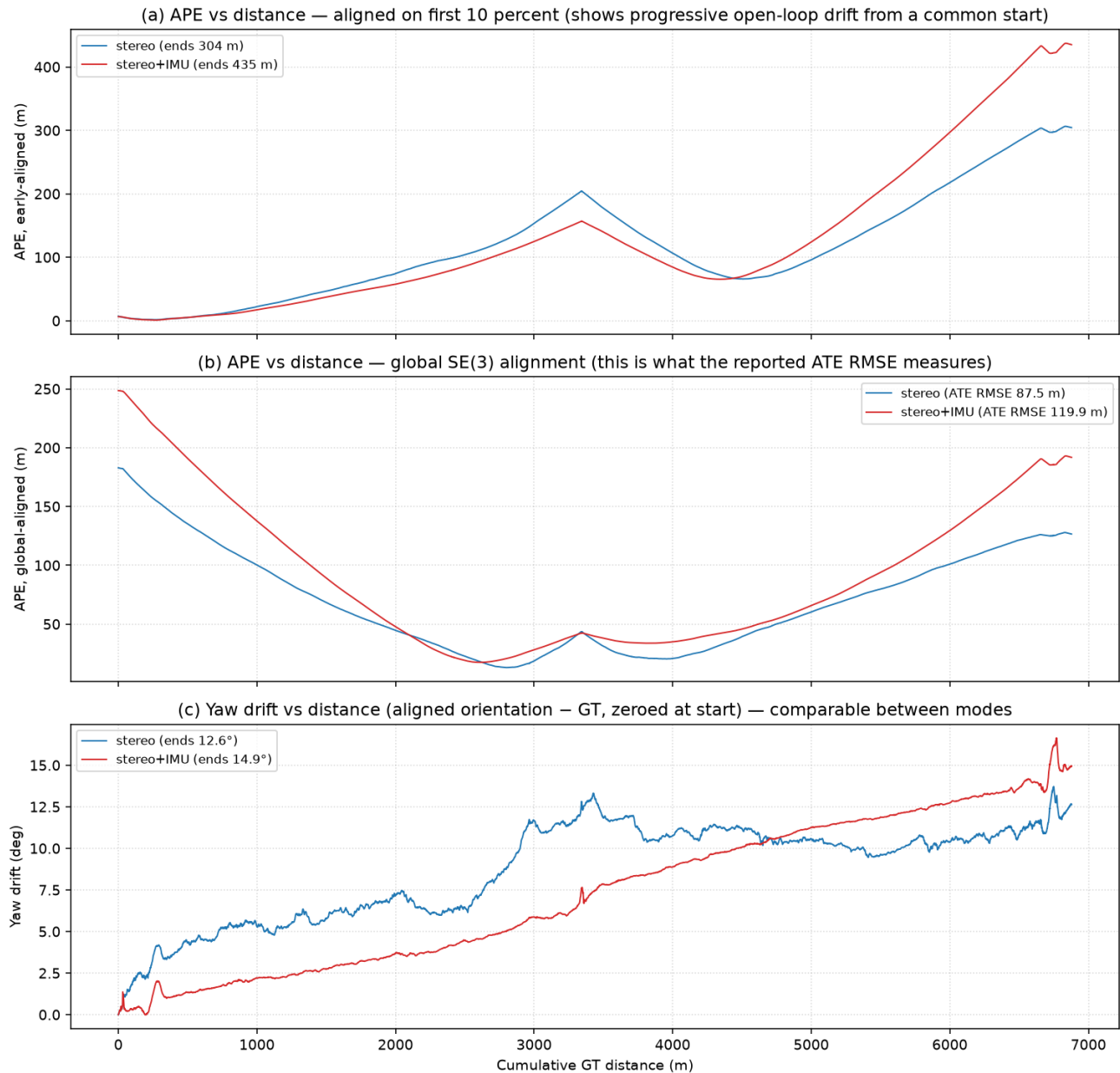
Dataset	License	Official/ commercial use	Practical config effort	Verdict
TartanDrive 2.0	MIT	allowed (attribution)	low — raw rectified stereo, repo calibration, ready odom	recommended official path
GO	CC BY-NC-SA 3.0	NonCommercial	high — compressed images, non-rectified stereo calib, NavSatFix→ENU	△ reference-only (clear w/ PM/legal)
4Seasons	CC BY-NC-SA (typ.)	NonCommercial	medium — fisheye + undistorted pinhole variants	reference-only, low novelty
KITTI	CC BY-NC-SA 3.0	NonCommercial	low (Odometry) / high (Raw VIO)	reference-only; Odometry no IMU, Raw stereo+IMU diverges
CADC	CC BY-NC 4.0	NonCommercial	medium-high — no standard stereo pair (picked 2 of 8 ring cams), inconsistent stereo calib, no raw IMU	reference-only; stereo shape good but metric scale off

Only **TartanDrive 2.0** is license-clean for product-facing work. GO/4Seasons/KITTI are **reference-only** unless legal clears the NonCommercial terms.

Known limitations (documented, not hidden)

1. **KITTI Odometry has no IMU — but KITTI VIO is not impossible; we ran it (KITTI Raw) and it diverged.** The Odometry benchmark ships stereo + LiDAR only, so its stereo+IMU cells are N/A. We went further and ran the separate **KITTI Raw** (OXTS) with a full stereo+IMU pipeline — pure stereo+IMU **diverges** (low excitation), and GPS-aided fusion is circular. Full root-cause in §10. (An earlier draft called KITTI+IMU "impossible" — corrected.)
2. **GO calibration is ambiguous.** The in-bag `camera_info` P-matrices are **not jointly stereo-rectified** ($f_x \neq f_y$, no baseline term). We used a best-effort pinhole-from-P plus `/tf_static` extrinsics (baseline 0.53 m). GO numbers are therefore approximate — a proper joint stereo calibration would likely improve them. GO is also CC BY-NC-SA (reference-only).
3. **GO full vs pilot differ a lot — and that is expected.** The 60 s pilot gave 1.60 m / 1.35 m; the **full 285 s / 865 m** run gives **11.66 m / 8.01 m**. Longer open-loop path → more accumulated drift. Both are real; the full-length number is the honest headline. (Drift% only rises from ~0.8 % to ~1.3 %, so the system is consistent; the metre figure grows with distance.)
4. **TD2 stereo+IMU is fragile.** On the slow 80 m turnpike pilot the IMU merely hurt (0.47→0.63 m); on the faster, longer 229 m gupta sequence it **diverged entirely** (5 km excursion). Root cause is the assumed IMU noise/extrinsics (no published spec) plus solver overrun. TD2 **stereo-only** is robust on both (0.47 m, 3.77 m). Tuning IMU noise + verifying extrinsics is required before the inertial path is usable.
5. **TD2 ground-truth caveat.** The earlier turnpike `/odometry/filtered_odom` + `gps.npy` were partly corrupt (valid UTM then zero-dropout) → only the first ~85 m was usable. **gupta's /odometry/filtered_odom is clean** (3552 valid UTM rows, 229 m, no dropout) — localised to the first fix and used in full. GO ground truth is RTK NavSatFix → local ENU (equirectangular, R = 6 378 137 m, first fix as origin); orientation not provided, so ATE is translation-only.
6. **Why drift% is the fair metric.** ATE in metres scales with path length: 119.93 m on 4Seasons office (6.9 km) is better relative accuracy than 0.63 m on an 80 m TD2 pilot. Only ATE ÷ distance compares an 80 m off-road clip to a 6.9 km suburban loop. All cross-dataset claims here use drift%.
7. **Pilots are short.** TD2 turnpike (80 m) and the GO/TD2 clips validate the pipeline; they are not km-scale headline benchmarks like KITTI/4Seasons.
8. **4Seasons office — IMU regression investigated, mechanism not conclusively isolated.** Stereo+IMU's ATE is 37 % higher than stereo-only's (119.93 vs 87.54 m) on this 6.9 km open-loop route. We tested the earlier claim that the IMU "amplifies yaw drift" by aligning both estimates to GT (SE(3) Umeyama, no scale) and plotting per-pose position error and yaw drift versus distance (figure below). What the data actually shows: - **It is progressive open-loop drift, not a constant offset.** Under early-window alignment both modes grow from ~6 m to hundreds of metres (stereo → ~304 m, stereo+IMU → ~435 m by route end), so the error is not a fixed bias present from the start. - **Yaw drift is comparable between the two modes** (~12–15° total, ≈2°/km). The IMU actually reduces yaw drift over the first half of the route and only exceeds stereo-only in the final third (≈15° vs ≈12.6° at the end) — the same region where the position-error gap opens. Late-route yaw drift therefore correlates with the gap but is far too small (an ~18 % yaw difference) to be the sole cause of a 37 % ATE difference. - **IMU bias / re-initialisation events could not be checked** — the prior stereo+IMU office run retained only its trajectory CSV, not its VINS stdout log.

Honest conclusion: the +37 % regression is real, but the root cause is **not conclusively isolated**. Candidate contributors include yaw drift, IMU extrinsic calibration, scale/translation drift, and initialisation quality. The original "inertial prior amplifies yaw drift" wording asserted a specific mechanism the evidence does not support, and has been softened throughout.



- GO full-285 s run-to-run stability — the single-run numbers are representative.** The headline GO numbers came from one run each, so we repeated both modes with the identical config and command (`go_rerun.sh`) and re-evaluated identically (`evo_ape --align --t_max_diff 0.05`):

Mode	Run 1 ATE RMSE	Run 2 ATE RMSE	Variation
GO stereo	11.6648 m	11.6588 m	0.05 % (0.006 m)
GO stereo+IMU	8.0064 m	8.0064 m	0.00 % (bit-identical)

Variation is negligible ($< 0.1\%$), as expected for single-threaded (`multiple_thread:0`) offline replay, which should be deterministic. Stereo+IMU reproduced bit-for-bit; the stereo run differed by one trajectory pose (8550 vs 8551, a frame-boundary timing artifact at bag start/end) for a 0.006 m ATE change. **The original GO numbers are confirmed representative.**

TD2 gupta divergence — root cause summary

In **stereo+IMU** mode VINS initialised successfully (online time-offset estimation, gyroscope bias calibrated to $(-0.008, 0.010, -0.009)$, "Initialization finish!") and then **diverged to a multi-kilometre position estimate** — the final pose is $(-5112, -6105, -2253)$ m on a route only 229 m long (ATE RMSE 2338 m). The most likely cause is a **mismatched IMU noise model combined with solver starvation**: TD2 publishes no IMU noise spec, so `acc_n/gyr_n/acc_w/gyr_w` are EuRoC-class defaults rather than values measured for TD2's actual Novatel SPAN unit, and the optimiser ran far over budget — **median 3.2 s / mean 4.4 s per frame against the 0.5 s `max_solver_time` budget, with 99 % of frames (676/680) over budget and a 41.8 s peak** — so the sliding-window estimate could not keep up and degenerated. Fixing it would require measuring the real Novatel noise densities (or empirically tuning `acc_n/gyr_n/acc_w/gyr_w` against a few known-good short sequences) and verifying the cam-IMU

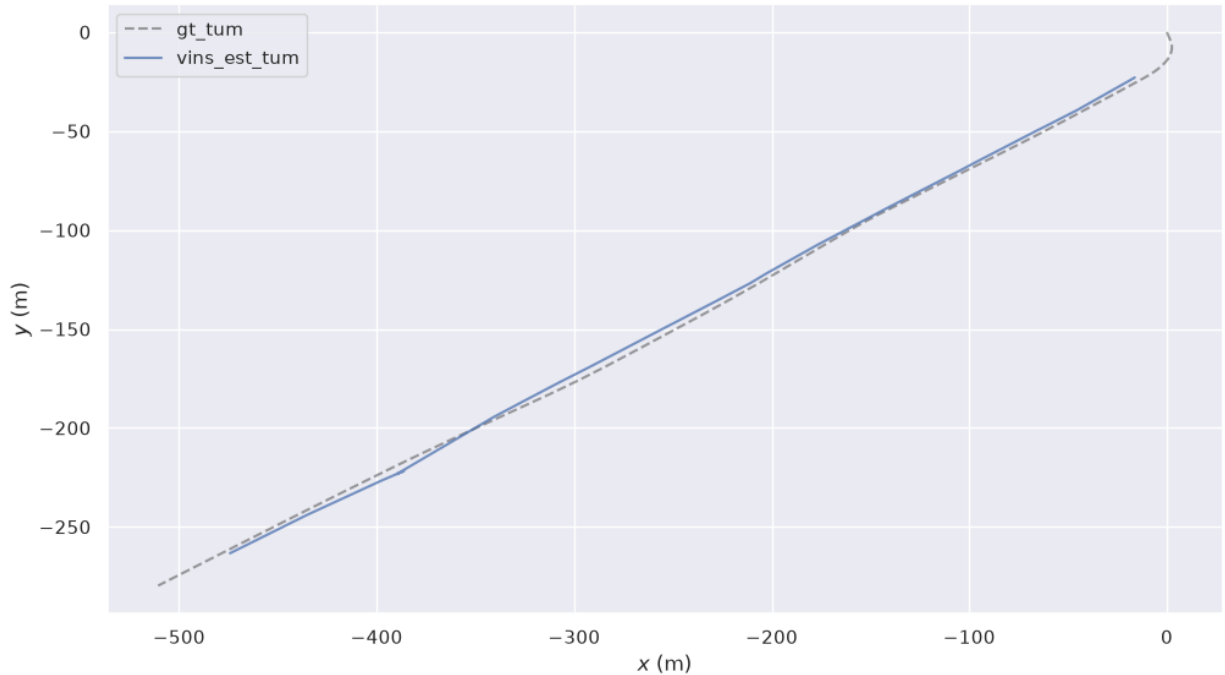
extrinsics. **This is isolated to stereo+IMU mode — stereo-only on the same gupta bag was stable (ATE 3.77 m, 1.65 % drift)** — so the failure is specifically in how the IMU factor interacts with this dataset's assumed calibration, not a general TD2 problem.

KITTI Raw stereo+IMU divergence — root cause summary

1. To fill the KITTI+IMU gap honestly we obtained **KITTI Raw 2011_10_03_drive_0027** (= Odometry seq 00), built a stereo+IMU bag from the **synced rectified stereo (10 Hz) + extract OXTS IMU (100 Hz)**, computed extrinsics from the calibration chain (`R_rect_00·R_velo_to_cam·R_imu_to_velo`; camera 1.10 m forward / 0.75 m up of the IMU), and ran `vins_node (imu:1)`. **It diverged**. After fixing two real issues — an initial 1.9 s gap in the OXTS stream (one isolated sample, then 100 Hz; we trimmed past it with `--start 3.0`) and confirming the bag is time-ordered — **VINS initialised cleanly (avg accel 0.52, 0.22, 10.02; gravity present) and tracked sanely for ~30 s / ~300 m** (x: 9.5→37.8→59.4→68→100 m), then **diverged catastrophically** (final pose $\sim 10^{26}$ m), with solver time ballooning to **4-76 s/frame** (budget 0.5 s). `estimate_td:0` did not help. The IMU data (accel includes gravity, $au \approx 9.8$), extrinsics (verified geometry), and bag ordering were all checked and are correct — so this is **not a pipeline bug**. The cause is **low IMU excitation**: KITTI's smooth, planar, near-constant-velocity driving makes IMU scale, gravity direction, and biases weakly observable, so the inertial estimate is under-constrained and corrupts over time (the ballooning solver time is the ill-conditioning showing through). This matches the VINS-Fusion authors' own choice to ship the KITTI-raw config with `imu:0` and run `kitti_gps_test` (stereo + GPS, IMU unused). **Stereo-only on the same data is fine (14.12 m, 0.38 %)**. The only "good" KITTI inertial number, 0.94 m, comes from GPS global fusion and is **circular** (scored against the same OXTS GPS it fuses) — reported for completeness, not as independent VIO accuracy. Honest verdict: pure stereo+IMU is **not usable** on this KITTI sequence; we report it as a divergence, not a number.

CADC — stereo works in shape but not in metric scale; no raw IMU (likely first VINS result on CADC)

1. **CADC (Canadian Adverse Driving Conditions, CC BY-NC 4.0 — reference-only)**. We ran drive `2018_03_06/0001` (543 stereo frames @ 10 Hz, 54 s, 592 m). We believe this is the **first published VINS-Fusion result on CADC**. Three things had to be worked out honestly:
 - **No purpose-built stereo pair**. CADC has 8 Ximea cameras in a surround ring. We selected the two forward cameras **cam00 (F) + cam01 (FR)**, which the calibration shows are near-parallel (relative rotation 1.59°) with a 0.512 m baseline; the other adjacent pairs rotate 43–74° (true ring cams). Visually confirmed both look forward with large overlap.
 - **Inconsistent stereo calibration → metric-scale error**. The trajectory shape is good — the scale-corrected (Sim(3)) ATE is **14.5 m (2.4 %)** with RPE 2.67 m, and the estimate overlays GT tightly (below). But under the standard **no-scale** metric the estimate is **1.76× too large** (ATE 142 m, 24 %). CADC's stereo extrinsic is internally inconsistent: baseline **0.512 m** (direct `T_00CAMERA_01CAMERA`), **0.399 m** (lidar-chain), and **~0.291 m** (scale-consistent with GT) all disagree, and **no provided value gives correct metric scale** (the 0.399 m run was worse: 190 m, 2.05×). So the 24 % is a **calibration artifact, not a VINS failure** — CADC's front cameras are not a cleanly-calibrated metric stereo rig. This is a stronger version of the GO calibration caveat.
 - **No raw IMU → stereo+IMU N/A**. The public CADC release exposes only the ~10–20 Hz Novatel INSPVAX fused pose (which we use as GT); there is no accelerometer/gyroscope stream. Deriving IMU from poses would be fabrication and 10–20 Hz is too slow anyway, so pure stereo+IMU cannot be run (parallel to KITTI Odometry). Adverse-weather note: images are overcast/wet winter scenes.



Recommendation

- **Adopt TartanDrive 2.0 (MIT), stereo-only, as the official off-road benchmark.** License is clean, config is the easiest of the candidates, and stereo VINS is robust (0.47 m / 0.59 % on turnpike, 3.77 m / 1.65 % on gupta).
- **Do not ship the TD2 stereo+IMU path yet.** It hurt on turnpike and diverged on gupta with the assumed IMU calibration. Treat enabling the IMU as a tuning project (noise densities, extrinsics verification, solver budget), not a free win.
- **Keep GO, 4Seasons and CADC as reference-only** (NonCommercial). GO and **CADC** additionally need a proper joint stereo calibration before their metric numbers are quotable — CADC's stereo shape is good but its metric scale is off by 1.76× (inconsistent extrinsics), and it has no raw IMU (§11).
- **Use KITTI stereo-only.** Odometry ships no IMU; KITTI Raw has OXTS but pure stereo+IMU **diverges** on it (low excitation, §10). KITTI remains the cleanest long-range stereo reference (0.27 %–0.39 % drift) — do not pursue KITTI VIO without addressing the excitation problem (e.g. sequences with more aggressive motion, or GPS global fusion, which is not pure VIO).
- **Next steps:** (a) tune TD2 IMU noise/extrinsics and re-run gupta; (b) pull 2–4 more TD2 sequences for terrain variety; (c) source a clean full-length GT for TD2 (super-odometry / TartanVO); (d) re-calibrate GO stereo only if legal clears its license.

Appendix — provenance (every number is reproducible)

Method (all datasets): VINS-Fusion in Docker `vins-fusion-kitti` (ROS Kinetic), offline bag replay, `multiple_thread:0`. Metrics via `evo_ape/evo_rpe`, `--align` (SE(3) Umeyama, no scale), TUM format. `vio.csv` → TUM: `ts/1e9 x y z qx qy qz qw`.

Result	Path
4Seasons stereo (this study)	<code>run logs ~/datasets/4seasons/output_stereo_{office,snow}/. neighbor ~/datasets/4seasons/run_4s_stereo.log; config ~/datasets/4seasons/stereo_cfg/4seasons_stereo.yaml; GT ~/datasets/4seasons/ground_truth/gt_{neighbor,office,snow}.txt</code>
4Seasons stereo+IMU (prior)	<code>~/github/VINS-Fusion/results/README.md, results/4seasons/</code>
KITTI stereo (Odometry)	<code>~/datasets/kitti/tuning_experiments.md</code> (tuned Run 3 + loop-closure runs)
KITTI Raw (2011_10_03_drive_0027 = seq 00)	<code>data ~/datasets/kitti_raw/2011_10_03/ (sync + extract OXTS); GT ~/datasets/kitti_raw/output/gt_kitti_10_03.txt (3723 m). Configs config/kitti_raw/kitti_10_03_stereo_imu.yaml (pure VIO) + kitti_10_03_gps_config.yaml (GPS). Runs ~/datasets/kitti_raw/output/{gps_run,imu_run}/ — vio.txt/ vio_global.csv/vio.csv, vins.log, eval_summary.txt. Scripts /tmp/{build_kitti_imu_bag,kitti_oxts_gt,vio_txt_to_csv}.py</code>
GO full 285 s	<code>~/Dataset_VINS_Fusion_Comparison_Project/results/go_vins_fusion/2025-01-24-13-07-50_newcal/{stereo_full,stereo_imu_full}/ — vio.csv, vins_est_tum.txt, reference_enu_tum.txt, go_full_eval_summary.txt; plots in results/comparison/figures/go_full*_traj_trajectories.png</code>
TD2 gupta	<code>~/Dataset_VINS_Fusion_Comparison_Project/results/tartandrive2_vins_fusion/2023-11-14-14-24-21_gupta/{stereo,stereo_imu}/ — vio.csv, vins_est_tum.txt, reference_odom_tum.txt, gupta_eval_summary.txt; plots results/comparison/figures/gupta*_traj_trajectories.png</code>

Result	Path
TD2 turnpike (pilot)	~/Dataset_VINS_Fusion_Comparison_Project/results/tartandrive2_vins_fusion/turnpike_warehouse/{stereo,stereo_imu}/
GO / TD2 configs	~/Dataset_VINS_Fusion_Comparison_Project/configs/{go,tartandrive2}_vins_fusion/
CADC (2018_03_06/0001) △NC	data ~/datasets/cadc/{raw,processed_raw}/ (cam00/01 @10Hz + novatel); calib ~/datasets/cadc/calib_probe/calib/; GT ~/datasets/cadc/ground_truth/gt_tum.txt (592 m); configs config/cadc/{cam0,cam1,cadc_stereo,cadc_stereo_imu}.yaml; run ~/datasets/cadc/output/stereo/ (vio.csv, vins_est_tum.txt, cadc_eval_summary.txt); bag builder ~/datasets/cadc/build_cadc_bag.py; devkit ~/datasets/cadc/cadc_devkit/; notes cadc_structure_notes.txt; plot results/figures/cadc_stereo_traj_trajectories.png
This report's chart	results/figures/drift_comparison.png (/tmp/drift_chart.py)
Office IMU investigation (§8)	results/figures/4seasons_office_imu_investigation.png (/tmp/office_investigate.py, /tmp/office_growth.py); inputs ~/datasets/4seasons/output_stereo_office/vio.csv (stereo) and ~/datasets/4seasons/output/vio_4seasons_office.csv (stereo+IMU) vs gt_office.txt
GO full-285 s run-to-run (§9)	~/go_vins_fusion/2025-01-24-13-07-50_newcal/{stereo_full_run2,stereo_imu_full_run2}/ (go_rerun.sh)

Raw evo figures retained per run: GO stereo_full ATE mean/median/max = 10.31 / 10.38 / 23.98 m (std 5.47); stereo_imu_full = 7.11 / 6.45 / 17.46 m (std 3.69). gupta stereo ATE mean/median/max = 2.96 / 2.10 / 9.60 m (std 2.33), RPE RMSE 0.67 m. 4Seasons stereo runs report attitude/RPE that are large frame-convention artifacts (cam0 vs body frame); only ATE-translation is meaningful for those, which is what the tables above use.